

# HB-PINNs: A Heat-Balance Physics-Informed Framework for Building Energy Loading Forecasting with Limited Data

Meng Zhang<sup>a</sup>, Zhihui Li<sup>b,\*</sup> and Zhibin Yu<sup>c,\*</sup>

<sup>a</sup>Mechanical and Aerospace Engineering, School of Engineering, University of Liverpool, Liverpool, L69 7ZX, UK

<sup>b</sup>Mechanical and Aerospace Engineering, School of Engineering, University of Liverpool, Liverpool, L69 7ZX, UK

<sup>c</sup>Mechanical and Aerospace Engineering, School of Engineering, University of Liverpool, Liverpool, L69 7ZX, UK

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## ABSTRACT

In the context of global decarbonization, buildings have received increasing attention due to their significant share in total energy consumption. Accurate prediction of building energy demand and indoor temperature forms the basis for effective control, demand response, and integrated energy management. Physics-informed neural networks (PINNs) offer a promising way to combine physical knowledge with data-driven learning. However, most existing work in the building domain focuses on subsystem-level tasks or control applications, with limited efforts directed toward whole-building performance prediction. In the limited studies that do consider whole-building applications, simplified resistance capacitance (RC) models are almost exclusively used as the physical constraint, since they are easy to formulate and computationally efficient. However, conventional RC models rely on simplified thermal representations and cannot explicitly represent some latent thermal processes, which may limit their prediction accuracy under varying operating conditions. In this study, a new method called heat-balance PINNs (HB-PINNs) is introduced. It directly uses the formula of heat balance to find out the latent physical quantities that can't be measured easily. In this way, the model can reconstruct a complete physical representation, so that the heating demand can be calculated more accurately. Under identical input conditions, HB-PINNs is evaluated against an RC-based PINN. The evaluation is carried out across multiple test sets, various training data ratios, and different prediction horizons. The proposed HB-PINNs achieves an average  $R^2$  of 94.99% across all scenarios and training ratios, compared with 88.69% for RC-PINNs. The results show that HB-PINNs is more accurate, and it has stronger adaptability under different operating conditions. These findings show that HB-PINNs provides an extensible and physical solution for building energy consumption modelling.

## Nomenclature

### Temperature variables

|                 |                                                            |
|-----------------|------------------------------------------------------------|
| $T_t$           | Setpoint temperature (proxy for indoor temperature)        |
| $T_{t-1}$       | Indoor temperature at previous time step                   |
| $T_z$           | Indoor zone air temperature                                |
| $T_{Si}$        | Inner surface temperature of surface $i$                   |
| $T_{out}$       | Outdoor air temperature                                    |
| $T_{\infty}$    | Outdoor air temperature used in infiltration heat exchange |
| $\Delta T$      | Short-term temperature change ( $T_t - T_{t-1}$ )          |
| $\Delta T_{in}$ | Indoor-outdoor temperature difference ( $T_t - T_{out}$ )  |

### Heat flow variables

|                 |                                                         |
|-----------------|---------------------------------------------------------|
| $Q_{sys}$       | Heating demand provided by HVAC system                  |
| $\hat{Q}_{sys}$ | Reconstructed heating demand from heat balance equation |
| $Q_i$           | Internal or external heat gain source $i$               |
| $Q_t^{heat}$    | Heating input at time $t$                               |
| $Q_t^{irr}$     | Solar heat gains                                        |
| $Q_t^{rest}$    | Internal heat gains (occupants, equipment, lighting)    |
| $c_z$           | Thermal capacitance of zone air                         |
| $h_i$           | Convective heat transfer coefficient of surface $i$     |
| $A_i$           | Area of surface $i$                                     |
| $\dot{m}_{inf}$ | Infiltration mass flow rate                             |
| $c_p$           | Specific heat capacity of air                           |

### RC model parameters

|       |                                       |
|-------|---------------------------------------|
| $a$   | Heating gain coefficient in RC model  |
| $b$   | Heat loss coefficient in RC model     |
| $e_1$ | Solar gain coefficient in RC model    |
| $e_2$ | Internal gain coefficient in RC model |

## 1. Introduction

The global energy system is undergoing a great transition as countries aim to reduce carbon emissions while maintaining energy security [1]. Buildings play a central role in this transition. Their heating, ventilation, and air-conditioning (HVAC) systems account for a substantial portion of total energy use and the related greenhouse gas emissions [2]. Accurate prediction of building performance is essential for a wide range of operational needs. These include optimisation, demand-response scheduling, maintenance planning, and the design or retrofit of energy-efficient buildings. Traditionally, such predictions have been based on two main approaches: physics-based simulation and data-driven statistical [3]. Physics-based simulations rely on established thermodynamic and heat-transfer formulations to represent building behaviour. They provide a structured way to capture how heat moves through the building and how indoor conditions evolve. Their limitations lie in high computational cost, the need for detailed geometric and material inputs, and the difficulty of calibrating them with real operational data [4]. In contrast, data-driven models ranging

\*Corresponding authors

 meng.zhang@liverpool.ac.uk (M. Zhang);

zhihui.li@liverpool.ac.uk (Z. Li); zhibin.yu@liverpool.ac.uk (Z. Yu)

from simple regressions to deep neural network (NN) can capture behavioural patterns directly from historical data without requiring explicit physical information. Their accuracy, however, depends strongly on data availability and quality, and they often show limited robustness or physical consistency when applied outside the range of the training data [5].

Physics-informed neural networks (PINNs) have been widely applied in various scientific and engineering domains, demonstrating strong capability in incorporating physical laws into data-driven learning [6],[7],[8]. Recent developments in PINNs have also provided a promising alternative for building energy modelling [9]. By incorporating governing physical equations into the training objective, PINNs achieve data-efficient and physically consistent learning beyond what purely data-driven methods can offer [10]. This improves prediction robustness under limited or noisy data, a common challenge in real building applications. These characteristics make PINNs particularly suitable for modelling thermodynamic behaviours in buildings, where reliable physics is available, but measurements are often sparse [11], [12]. Existing PINNs studies in the building domain fall into three categories: subsystem-level modelling, control and demand response, and whole-building prediction. Subsystem-level work confirms the feasibility of embedding physics into neural networks but remains limited to isolated components. Studies on control and demand response improve robustness by incorporating physical constraints, but they generally focus on control strategies rather than on the accuracy of the underlying models. Whole building PINNs modelling remains rare, despite its importance for integrated operation and reliable prediction of indoor temperature and energy use.

Buildings provide a natural application area for PINNs, as their behaviour is strongly shaped by underlying physical laws. Owing to the complexity of whole-building systems, most existing studies have applied PINNs to specific subsystems such as wall heat transfer, airflow, or HVAC components, rather than to full building-scale models. For example, PINNs have been employed to integrate fundamental thermodynamic laws into neural network models for predicting heat pump loads [13]. Kang et al. [14] applied a PINNs to estimate the outlet water temperature in cooling towers of cooling systems, while Li et al. [15] developed a PINNs framework for predicting the outlet temperature of thermal energy storage systems in buildings. Jing et al. combined fluid dynamics with PINNs to rapidly predict indoor temperature, leveraging high-fidelity yet sparse data collected from CFD simulations and temperature sensors [16]. Fang et al. [17] demonstrated that incorporating physics-based constraints via PINNs can improve the prediction accuracy of chiller coefficient of performance (COP). Beyond thermal systems, Labib et al. [18] incorporated the Radiative Transfer Equation (RTE) and the diffusion equation into a machine learning framework. In the domain of ventilation, Son et al. [19] proposed a PINNs-based model to estimate ventilation and infiltration rates using long-term observational data.

Similarly, Wei et al. [20] applied a PINNs to reconstruct indoor airflow fields based on limited measurement data. Boussaid et al. developed a model that learns the thermo-hydraulic transients of district heating networks by modelling energy balance along pipelines [21]. Pandiyan et al. proposed a PINNs that embeds electric water heater thermal dynamics into the training process [22]. Beyond subsystem modelling, PINNs have also been explored in the context of building energy system control and demand response. Wang et al. proposed a physics-informed input convex neural network to predict indoor environmental dynamics and incorporated it into a hierarchical data-driven predictive control framework for HVAC systems [23]. Bunning et al. [24] showed that physics-informed regression models can outperform more complex machine learning approaches in model predictive control (MPC). Liang et al. [25] proposed a PINNs framework for chiller plant optimal control that incorporates both structure-type and trend-type prior knowledge. Pavirani et al. [26] investigated demand response for residential building heating by integrating PINNs into a Monte Carlo Tree Search (MCTS) control framework.

Another line of research that has not yet received as much attention is whole-building performance prediction, where the primary focus lies on improving the accuracy and robustness of prediction models themselves rather than control strategies or subsystem optimization. This kind of work emphasizes the development of physics-informed approaches that can capture the overall building dynamics and deliver reliable predictions of both energy consumption and indoor thermal conditions. Gokhale et al. developed a PINNs that embeds the RC thermal equations into the training process to predict room temperature, power consumption, and the thermal-mass state of a building [27]. Yang et al. [28] developed a physics-constrained graph neural network for zone temperature prediction, combining graph-based spatial representation with RC-model constraints. Ma et al. [29] proposed a physics-informed ensemble framework for predicting building heating and cooling loads. They further demonstrated that low-fidelity RC baselines are sufficient within this scheme, achieving accuracy comparable to high-fidelity *EnergyPlus* while offering higher efficiency. Chen et al. [30] proposed a PINNs combining a RC thermal model with data-driven learning, achieving more accurate and generalizable whole-building temperature and load predictions than purely data-driven models. Di Natale et al. [31] proposed a model enforce physical consistency through network architecture rather than residual-based training.

Several gaps remain in current research. Most PINNs studies focus on isolated subsystems, and work on control or demand response emphasises control performance rather than model accuracy. Whole-building PINNs modelling is still rare, despite its importance for reliable temperature and energy prediction. Simplified approaches such as resistance capacitance PINNs (RC-PINNs) appear in most existing studies. Key challenges include achieving sufficient accuracy for complex building systems, balancing model simplification with physical fidelity, and ensuring generalisation

when only limited data are available. To address these challenges, this study proposes the heat-balance PINNs (HB-PINNs) framework. Unlike simplified RC-based approaches, the method uses the PINNs structure to estimate physical variables in the heat-balance equation that cannot be directly measured, allowing a more complete physical formulation and enabling accurate reconstruction of heating demand.

The rest of the paper is structured as follows. Section 2 introduces the methodological framework and the procedures used to build the model. The results and the comparison between all model variants are reported in Section 3. Section 4 summarises the main findings.

## 2. Methodology

As shown in Fig. 1, the framework combines physics-based modelling with data-driven learning to estimate heating demand under a range of operating conditions. For first step, high-resolution hourly demand data are generated using *EnergyPlus* for a representative residential building. A single thermal zone is selected for detailed analysis. The obtained time series data are processed and then used to train and test the model. In order to add physical knowledge to the learning process, two forms of constraints are introduced. The first is a comprehensive heat balance model, which can capture the heat conduction, heat convection and heat radiation in the indoor environment. The second is a simplified RC model, which can simulate the heat transfer dynamics among outdoor air, wall layer and indoor space. The neural network itself follows a standard fully connected feedforward structure. In order to assess whether the model remains reliable when only limited data are available, the proportion of training data is gradually reduced from 100% (a full year with 8760 hourly samples) to 40% (less than half a year). For each data configuration, Bayesian optimisation is applied to adjust key parameters ensuring that the training process remains fair. In addition, its prediction performance is evaluated under five different heating-control scenarios: Eco (energy-saving priority), Comfort (comfort priority), Work Shift (two-stage control), Peak Only (extreme energy-saving mode) and Variable (fluctuation control). These test cases reflect a series of different operating preferences and dynamic thermal conditions.

### 2.1. Building Simulation

The development of data-driven heating-demand models relies on access to large volumes of high-quality time-series data. However, these data are not easy to obtain from real buildings. Various technologies are used in residential heating systems, such as gas boilers, heat pumps and district-heating units. And each system is operated with its own control logic and measurement method. Laboratory measurements can provide accurate data, but their scale is too small. Real-scene data are often noisy or incomplete. They may contain missing values or disturbances caused by uncontrollable factors such as user behaviour and window opening.

To address these issues, a physics-based building simulation is used to generate data. *EnergyPlus*, a mature building-simulation tool, is used in this study. The simulation framework is designed to allow full control of key inputs. Hourly heating demand and -operation data are produced through the simulation. The final dataset is complete, noise-free and reproducible. It supports fair comparison across different model configurations. It also allows controlled testing under different levels of data availability and heating-control scenarios.

#### 2.1.1. Building Constructions

The building used in this study is a typical single-storey UK residential prototype. It is modelled as a single thermal zone with a total floor area of about 18.9 m<sup>2</sup> and a volume of 56.8 m<sup>3</sup>. The envelope includes four exterior walls with areas between 10.5 and 24.3 m<sup>2</sup> and an average thermal transmittance of around 1.5 W/m<sup>2</sup>K, a flat roof of 23.3 m<sup>2</sup> with a U-value of 0.25 W/m<sup>2</sup>K, and a ground-contact floor of the same area with a U-value of 0.87 W/m<sup>2</sup>K. The building has four external windows with a combined area of 16.5 m<sup>2</sup>, giving a window-to-wall ratio of about 26%. The windows are double glazed with a 3 mm pane, a 13 mm air gap, and a 3 mm pane with a low-emissivity coating on the inner surface. The centre-of-glass U-value is around 2.7–3.0 W/m<sup>2</sup>K, but the overall heat loss is greater due to the relatively high frame transmittance of 9.5 W/m<sup>2</sup>K. The heating setpoint is kept at 18 °C for most of the day and reduced to 12 °C at night. No cooling system is installed, and space heating is provided by an ideal system. Internal conditions include a small number of occupants with a density of about 0.0155 per square meter, domestic hot water use, and air infiltration of roughly 0.30 air changes per hour. The simulation uses 2020 historical weather data, with outputs generated at a 60-minute timestep. The main output variable is the space heating demand which is used as the target for training and validating the neural network and PINNs.

#### 2.1.2. Heating Profile

To reflect the diversity of real-world heating behaviours, a set of distinct heating schedules was created within *EnergyPlus*. These schedules aim to capture variations not only in temperature setpoints but also in system operation timings. Both the target temperature and the on-off periods of the heating system vary across the profiles. A total of 5 heating strategies are implemented, as shown in Fig. 2. These include Eco, Comfort, WorkShift–Weekday, WorkShift–Weekend, PeakOnly, and Variable. Each represents a different usage pattern or control philosophy. Eco mode will use a relatively low base temperature, and the daily heating time is very short. Comfort mode is different. It keeps a constant temperature of 21°C all day. The two WorkShift modes distinguish between working days and weekends and adjust them according to time changes. The PeakOnly schedule is only heated in two short periods, morning and evening, which uses the least energy. Variable mode is even different, its temperature and time will change irregularly, just like

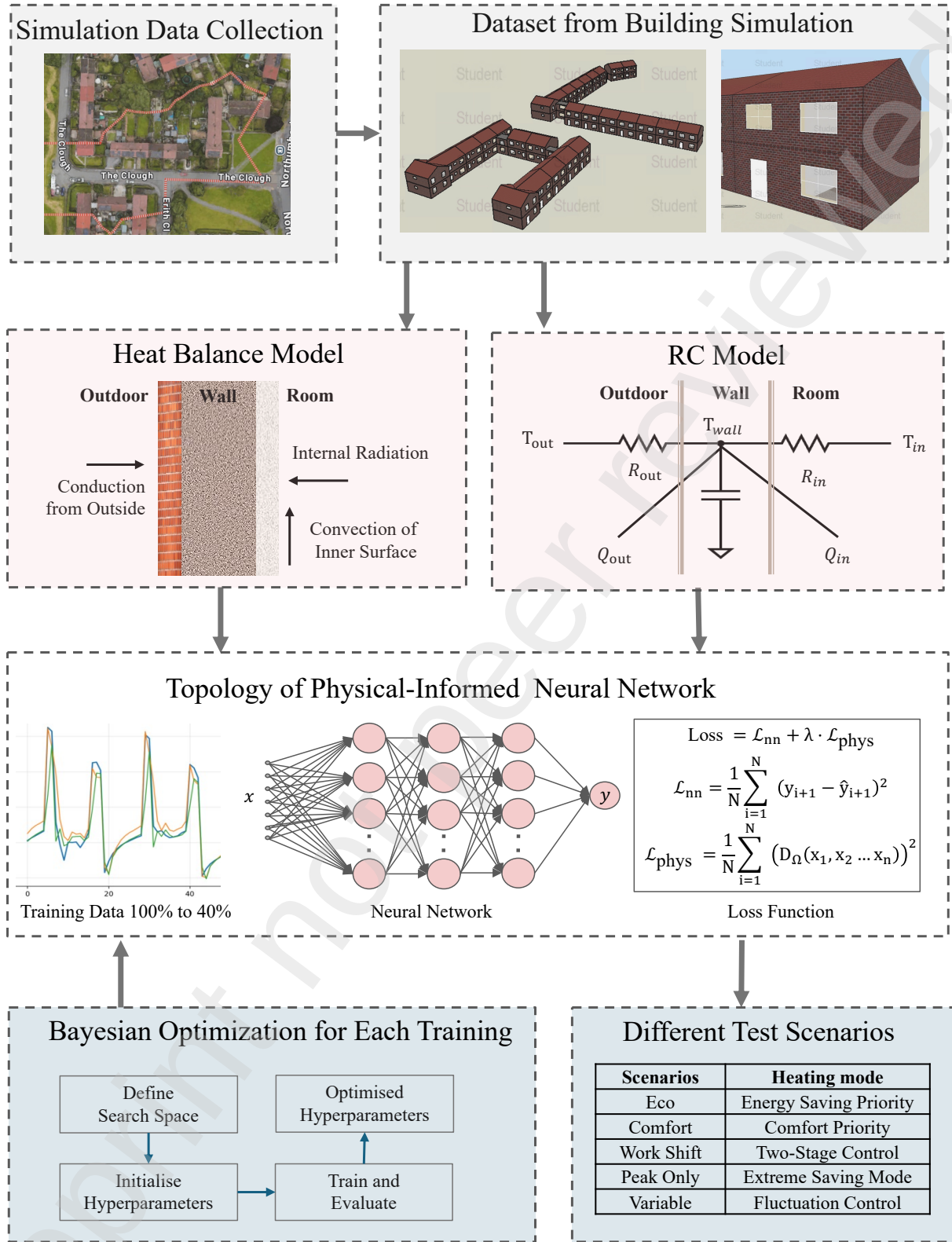


Figure 1: Block Diagram of the Model

the behavior of people in it can't be predicted. With these different control methods, the building has produced various heating needs. This design creates a set of operational

scenarios that test the model's ability to generalize beyond the specific dynamics observed during training.

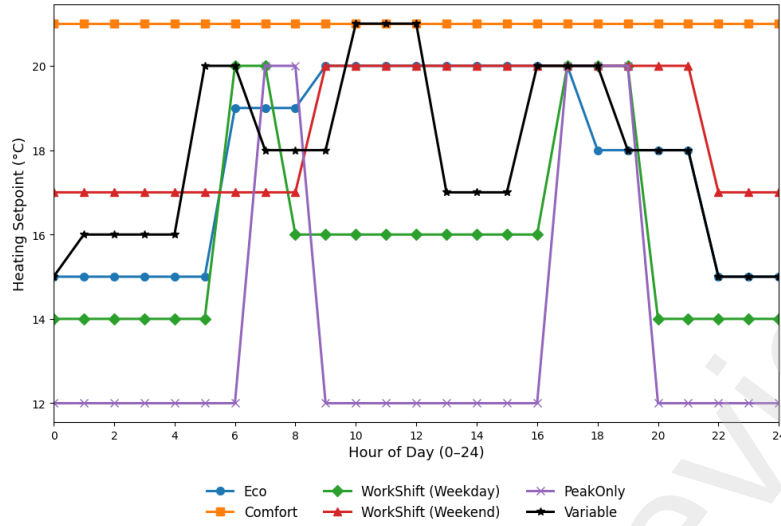


Figure 2: Different Heating Schedule for Testing

## 2.2. Physics-Based Prediction of Heating Demand

Although *EnergyPlus* provides highly detailed physics-based calculations involving thousands of equations, directly embedding the full set of governing equations into a PINNs framework would be computationally intractable and unnecessarily complex. Instead, two complementary physics-based approaches are employed to represent the building thermal behaviour at different levels of abstraction. The first is a zone-level heat balance model, derived from the fundamental principle of energy conservation, which provides a comprehensive accounting of thermal storage, surface heat exchange, and infiltration processes within the selected zone. The second is a reduced-order RC network model, which captures the dominant heat transfer. It can represent conduction through the envelope and convective exchange with indoor air by using a small number of lumped parameters. The RC model offers a computationally efficient representation that is well-suited for integration with data-driven models, allowing the PINNs to be trained under explicit physical constraints without requiring the full complexity of the *EnergyPlus* solver.

### 2.2.1. Heat Balance Constraint

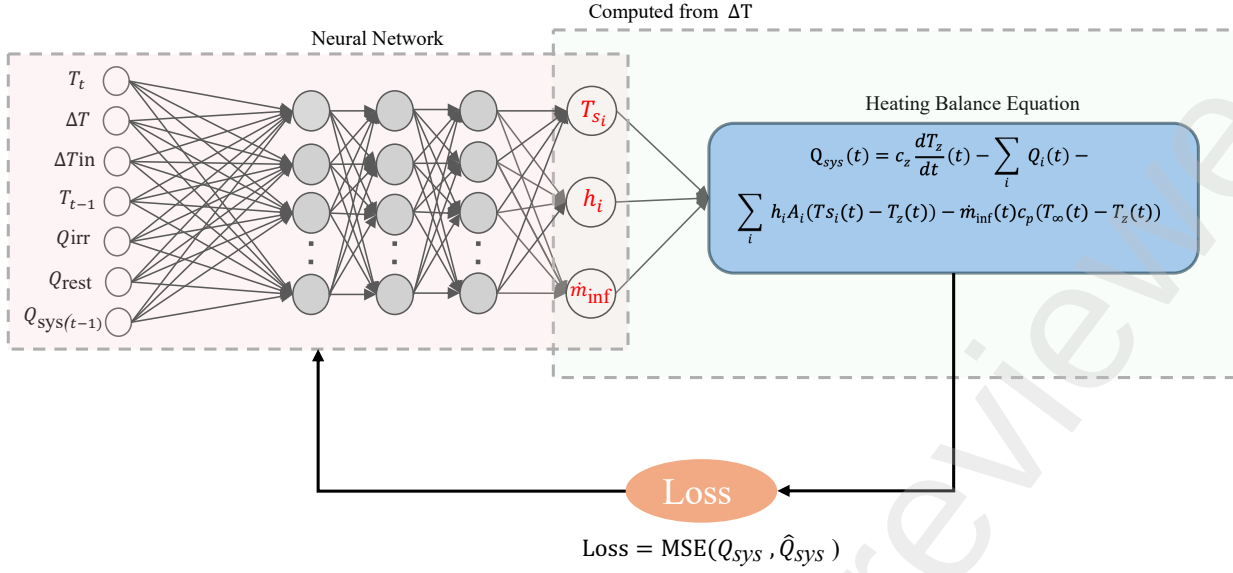
The heat balance model is based on the principle of energy conservation applied to a control volume representing the building zone. The internal energy variation of the zone air must equal the sum of all thermal exchanges with the surroundings and the load supplied by the HVAC system. The system load is thus determined by (i) transient thermal storage, (ii) internal and external gains, (iii) convective heat transfer with interior surfaces, and (iv) infiltration with outdoor air. This comprehensive accounting embeds the key physical processes that govern zone-level heating demand.

$$Q_{\text{sys}}(t) = c_z \frac{dT_z}{dt}(t) - \sum_i Q_i(t) - \sum_i h_i A_i (T_{Si}(t) - T_z(t)) - \dot{m}_{\text{inf}}(t) c_p (T_{\infty}(t) - T_z(t)) \quad (1)$$

Where:

- $Q_{\text{sys}}(t)$ : heating load provided by HVAC system
- $c_z \frac{dT_z}{dt}$ : change in internal energy of zone air
- $\sum_i Q_i(t)$ : internal and external heat gains (e.g., occupants, lighting)
- $\sum_i h_i A_i (T_{Si} - T_z)$ : convective heat transfer from surfaces
- $\dot{m}_{\text{inf}}(t) c_p (T_{\infty} - T_z)$ : sensible load from infiltration

This formulation ensures that the predicted heating demand reflects both the dynamic temperature response and all concurrent thermal interactions. It allows the learning process to be grounded in first-principles physics, rather than relying purely on empirical correlations. However, using this full heat balance equation directly within PINNs training presents two main challenges. First, it increases computational cost. Second, many key variables (such as surface temperatures, heat transfer coefficients, and infiltration rates) are not available in real-world datasets due to sensing limitations. To overcome this, the present study employs a physics-informed inverse formulation. Instead of directly using observed system loads to compute residuals, the PINNs is trained to infer latent physical variables from available measurements of zone temperature and total load.



**Figure 3:** HB-PINNs Architecture for Inverse Estimation of Physical Variables with Heat Balance Supervision

### 2.2.2. RC Constraint

The RC-based method describes indoor temperature dynamics using a simplified autoregressive formulation [31]:

$$T_{t+1} = T_t + aQ_t^{\text{heat}} - b(T_t - T_t^{\text{out}}) + e_1 Q_t^{\text{irr}} + e_2 Q_t^{\text{rest}} \quad (2)$$

This expression reflects the energy balance of the zone air, where the temperature change is influenced by heating input, solar gains, internal gains, and heat exchange with the outdoor environment and neighbouring zones. The coefficients  $a, b, e_1, e_2$  correspond to effective thermal capacitance, thermal resistances, and gain allocation factors, and can be identified from data or estimated from prior building information. This simplified structure is limited in its ability to capture coupled and time-varying heat transfer processes. These limitations motivate the development of a heat-balance PINNs framework that more explicitly represents physical interactions and latent thermal processes.

## 2.3. RC-PINNs and HB-PINNs

### 2.3.1. Topology of Model

The architecture of the HB-PINNs model is illustrated in Fig. 3. The network receives seven observable and thermally relevant features as input. The seven inputs are: (1) setpoint temperature  $T_t$  (used here as a proxy for room temperature); (2) short-term temperature change  $\Delta T = T_t - T_{t-1}$ ; (3) indoor–outdoor temperature difference  $\Delta T_{\text{in}} = T_t - T_{\text{out}}$ ; (4) solar gains  $Q_t^{\text{irr}}$ ; (5) internal gains  $Q_t^{\text{rest}}$ ; (6) heating demand at the previous step  $Q_{\text{sys}(t-1)}$ ; and (7) previous room temperature  $T_{t-1}$ .

Instead of directly predicting the heating demand, the HB-PINNs output a set of latent physical variables that are not observed during training, including the inner surface temperatures  $T_{s,i}$ , convective heat transfer coefficients  $h_i$ ,

mapped to surfaces, and the infiltration mass flow rate  $\dot{m}_{\text{inf}}$ . These quantities are not directly observed or labelled during training. Instead, they are inferred through supervision imposed by the governing physics.

At each training step, the network predicts the latent variables, which are then substituted into the heat-balance equation to compute a reconstructed heating load  $\hat{Q}_{\text{sys}}$ . The deviation between  $\hat{Q}_{\text{sys}}$  and the observed heating demand  $Q_{\text{sys}}$  defines a physics-based loss term, which is minimized alongside regularization terms during training. In practical applications,  $Q_{\text{sys}}$  can be obtained from HVAC operational measurements, such as heating coil energy consumption, or calculated from supply air temperature and flow rate using standard energy balance relations. In this study,  $Q_{\text{sys}}$  is obtained from *EnergyPlus* simulation outputs to provide a consistent reference for model training and evaluation. For RC-PINNs, given the same seven inputs, a feedforward neural network directly regresses  $Q_{\text{sys}}$ . Before training, an RC-based linear model is fitted on the training data. During training, the total loss jointly considers the NN loss and the physics-related loss derived from the RC model.

### 2.3.2. Bayesian Optimisation

Due to the distinct modelling structures of RC-PINNs and HB-PINNs, separate Bayesian optimisation strategies are employed to enhance model accuracy and enforce physical consistency. Both models utilise the *Optuna* framework to search for optimal hyperparameters, with the objective of minimising the physics-informed loss on the validation set. For RC-PINNs, this loss corresponds to the residual of the RC model, whereas for HB-PINNs it is derived from the heat-balance residual. In the RC-PINNs framework, the neural network directly predicts the heating demand, and a simple RC model is first fitted to the training data to

**Table 1**

Hyperparameter optimisation ranges and physical regularisation settings.

| Parameter                                   | Minimum    | Maximum    | Description                                                | Exclusive to |
|---------------------------------------------|------------|------------|------------------------------------------------------------|--------------|
| <b>Model hyperparameter optimisation</b>    |            |            |                                                            |              |
| Number of hidden layers                     | 1          | 5          | Number of hidden layers in the neural network              | Both         |
| Neurons in each layer                       | 10         | 200        | Number of neurons per hidden layer                         | Both         |
| <b>Training hyperparameter optimisation</b> |            |            |                                                            |              |
| Initial learning rate ( $\eta_0$ )          | 0.0001     | 0.01       | Starting learning rate before scheduling                   | Both         |
| Learning rate drop period (P)               | 50         | 200        | Epoch interval for applying learning rate decay            | Both         |
| Learning rate drop factor (F)               | 0.1        | 0.5        | Factor by which the learning rate is reduced               | Both         |
| Max epochs                                  | 50         | 200        | Maximum number of training epochs                          | Both         |
| Mini-batch size                             | 16         | 128        | Number of samples per training batch                       | Both         |
| Gradient threshold                          | 0.5        | 2          | Gradient clipping threshold to prevent exploding gradients | Both         |
| <b>Additional physical regularisation</b>   |            |            |                                                            |              |
| Physics loss weight ( $\lambda$ )           | 0.01       | 10         | Weight balancing physics-based and data-driven losses      | Both         |
| L2 weight decay                             | 1e-6       | 1e-2       | Regularization for model generalisation                    | Both         |
| Dropout rate                                | 0          | 0.5        | Dropout to prevent overfitting                             | Both         |
| Early stopping patience                     | 10         | 20         | Validation patience threshold                              | Both         |
| Learning rate scheduler type                | N/A        | N/A        | Type of learning rate schedule (cosine / step / warmup)    | Both         |
| Bounds                                      | Applicable | Applicable | Enforce physical bounds on $h_i$ and $\dot{m}_{inf}$       | HB-PINNs     |
| Temporal smoothness                         | Applicable | Applicable | Penalise abrupt variation of physical parameters           | HB-PINNs     |

serve as a physical reference. During training, the total loss combines the neural-network prediction error with the difference between the NN output and the RC-model output. This regularisation encourages the network to remain consistent with fundamental thermal behaviour represented by the RC dynamics.

In contrast, HB-PINNs infer a set of latent physical variables, including surface heat transfer coefficients and infiltration rates. These latent variables are substituted into the heat-balance equation to compute a reconstructed heating load. The training objective is to minimise the discrepancy between this reconstructed load and the observed heating demand, thereby enforcing a stronger physical consistency than RC-PINNs. Both RC-PINNs and HB-PINNs optimise a common set of hyperparameters, including network depth, hidden-layer width, activation function, learning rate, batch size, and regularisation strength, as summarised in Table 1. Additionally, HB-PINNs incorporate physical regularisation terms that constrain the latent thermal variables within realistic bounds and penalise abrupt temporal fluctuations. All models are trained with a warm-up learning-rate schedule and employ early stopping to improve robustness and generalisation.

### 2.3.3. Training and Testing Data

Both RC-PINNs and HB-PINNs are trained and evaluated using the same time-series dataset collected from a representative residential building zone. The training dataset

is drawn from a designated reference scenario, and five proportions of the training data (100%, 80%, 60%, 40%, and 20%) are considered to assess the robustness of each model under different levels of data availability. This allows assessment of model robustness under varying data availability. To evaluate generalization performance, testing is conducted on five scenarios, each representing a distinct boundary condition or operational pattern. All input features are normalized based on the training set statistics. All machine-learning model training and evaluations are implemented in *Python* using the *PyTorch* framework. The models are executed on an HP Z2 Tower G9 workstation equipped with an *Intel Core i9-14900K* processor, 128 GB RAM, and multiple GPUs providing a total of 24 GB VRAM, running a 64-bit *Windows* operating system.

## 3. Results and Analysis

This section presents the evaluation results of the proposed modelling framework. Section 3.1 introduces the reference heating demand profiles generated by *EnergyPlus*, which serve as the ground truth for model training and testing. Based on these benchmark profiles, Section 3.2 provides an overview of model performance across four representative approaches: the RC model, a purely data-driven NN, RC-PINNs and HB-PINNs. Section 3.3 compares the performance of RC-PINNs and HB-PINNs under different data volumes and test conditions, focusing on their generalization ability and robustness. Section 3.4 further evaluates

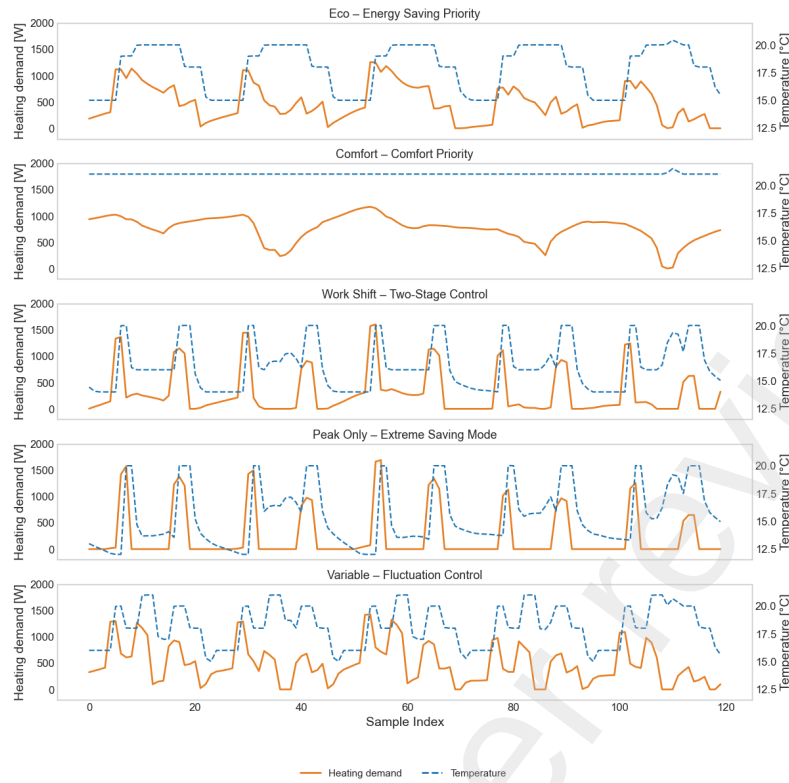


Figure 4: Heating Profile for Heating Demand and Corresponding Air Temperature

the multi-step prediction ability of these two PINNs models, and especially explains their performance in long-term prediction.

### 3.1. Reference Heating Demand Profiles from Simulation

As mentioned, five representative heating control scenarios are used to simulate five representative heating control scenarios. Each scenario corresponds to different control logic, which represents various common user preferences in intelligent buildings. Fig. 4 shows the heating demand and indoor temperature changes in these modes. In all cases, there is an obvious time relationship between heating demand and indoor air temperature. The change of heating demand always occurs earlier than the change of temperature. This situation shows that the control system will be prepared in advance, and the building itself has thermal inertia. The heating-demand profile is not determined by the setpoint alone. Even in the Comfort scenario, where the setpoint remains fixed, the heating demand still shows small fluctuations. In contrast, the heating demand of Eco and Peak Only modes are intermittent. These two curves are typical representatives of those control strategies that "prefer to save electricity even if the temperature fluctuates a little". The time changes in the Variable and Work Shift modes are more complicated. Their demand is adjusted throughout the day based on preset arrangements or external conditions. These situations bring additional challenges to accurately forecast the demand.

### 3.2. Overview of Model Performance

Before RC-PINNs and HB-PINNs are compared in detail, it needs to be checked whether PINNs perform better than traditional modelling methods. If they are not as effective as basic approaches such as the RC model or a data-only neural network, then using more complex PINNs has little practical value. For this reason, four modelling methods (RC, NN, RC-PINNs and HB-PINNs) are compared under five heating-control situations and different training ratios. The results provide a foundation for the focused comparison between RC-PINNs and HB-PINNs in Sections 3.3 and 3.4.

Table 2 reports the prediction accuracy ( $R^2$ ) of all four models at four training ratios (100%, 80%, 60%, and 40%). The coefficient of determination reflects how accurately the predicted heating demand explains the variations in the real data. In all cases, RC-PINNs achieve higher accuracy than the pure NN model, showing the benefit of adding the RC constraint. This difference becomes particularly clear when the amount of training data is limited. For example, in the Comfort scenario with only 40% of the data used for training, the NN model drops to negative accuracy ( $R^2 = -13.7\%$ ), while RC-PINNs maintain a high accuracy of 93.5%. A similar pattern is seen in dynamic scenarios such as Peak Only and Variable: the NN model struggles to exceed 60%, whereas RC-PINNs remain above 77% and 88%, respectively. Even under more stable strategies, such as Eco and Work Shift, RC-PINNs show better stability and higher accuracy than the NN model.

**Table 2**Prediction performance ( $R^2$ ) of NN, RC, RC-PINNs, and HB-PINNs across five test scenarios and four training ratios.

| Train %           | Test Set | NN      | RC            | RC-PINNs      | HB-PINNs      |
|-------------------|----------|---------|---------------|---------------|---------------|
| <b>Comfort</b>    |          |         |               |               |               |
| 100               |          | 90.79%  | 97.01%        | 96.83%        | <b>97.80%</b> |
| 80                |          | 86.04%  | <b>97.33%</b> | 91.03%        | 96.80%        |
| 60                |          | 93.33%  | <b>97.16%</b> | 95.38%        | 94.90%        |
| 40                |          | -13.70% | 96.59%        | 93.53%        | <b>98.50%</b> |
| <b>Eco</b>        |          |         |               |               |               |
| 100               |          | 91.58%  | 94.47%        | 96.29%        | <b>98.30%</b> |
| 80                |          | 84.87%  | 93.94%        | 92.38%        | <b>97.30%</b> |
| 60                |          | 89.65%  | 94.28%        | 94.63%        | <b>96.90%</b> |
| 40                |          | 54.94%  | 94.29%        | 92.99%        | <b>98.50%</b> |
| <b>Peak Only</b>  |          |         |               |               |               |
| 100               |          | 68.44%  | 81.32%        | <b>91.56%</b> | 88.10%        |
| 80                |          | 54.91%  | 75.35%        | 67.86%        | <b>85.30%</b> |
| 60                |          | 57.63%  | 78.84%        | 74.84%        | <b>90.30%</b> |
| 40                |          | 54.13%  | 83.60%        | 77.77%        | <b>91.30%</b> |
| <b>Variable</b>   |          |         |               |               |               |
| 100               |          | 81.58%  | 91.04%        | 93.65%        | <b>96.80%</b> |
| 80                |          | 74.59%  | 88.89%        | 83.93%        | <b>95.20%</b> |
| 60                |          | 79.14%  | 90.18%        | 90.82%        | <b>95.30%</b> |
| 40                |          | 46.53%  | 91.64%        | 88.56%        | <b>97.40%</b> |
| <b>Work Shift</b> |          |         |               |               |               |
| 100               |          | 81.79%  | 89.10%        | 94.21%        | <b>95.50%</b> |
| 80                |          | 73.33%  | 86.31%        | 82.48%        | <b>93.70%</b> |
| 60                |          | 76.79%  | 87.96%        | 88.03%        | <b>95.10%</b> |
| 40                |          | 51.99%  | 89.93%        | 86.94%        | <b>96.90%</b> |

However, the performance of RC-PINNs is usually similar to that of RC model alone, which shows that the accuracy of prediction is basically limited by the simple physical structure of RC model. Because RC model simplifies the thermal system of a building into total resistances and capacitances, it lacks the ability to capture complex thermal processes. This limitation becomes more obvious when the boundary conditions change greatly or less training data available.

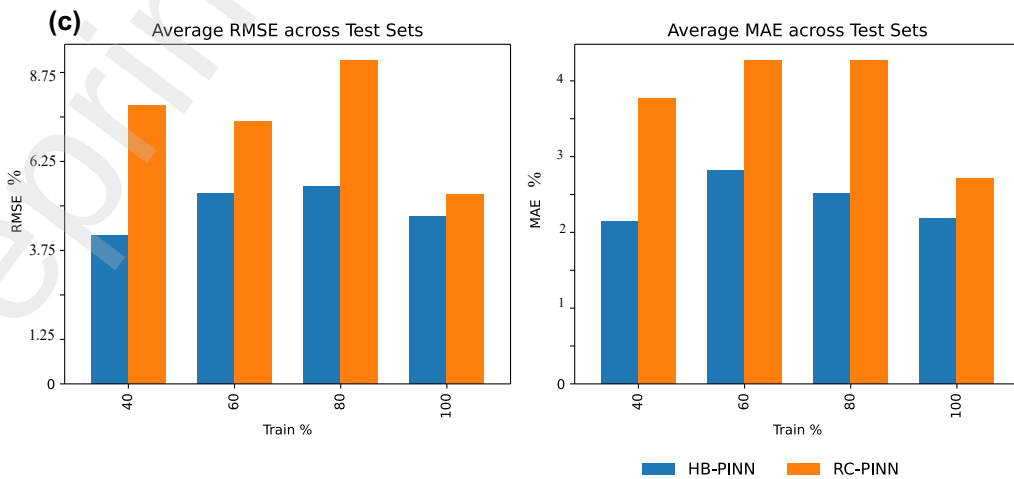
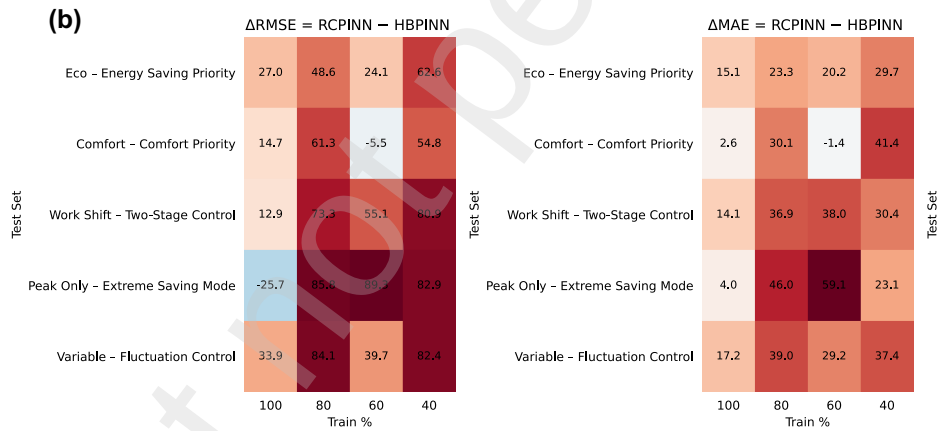
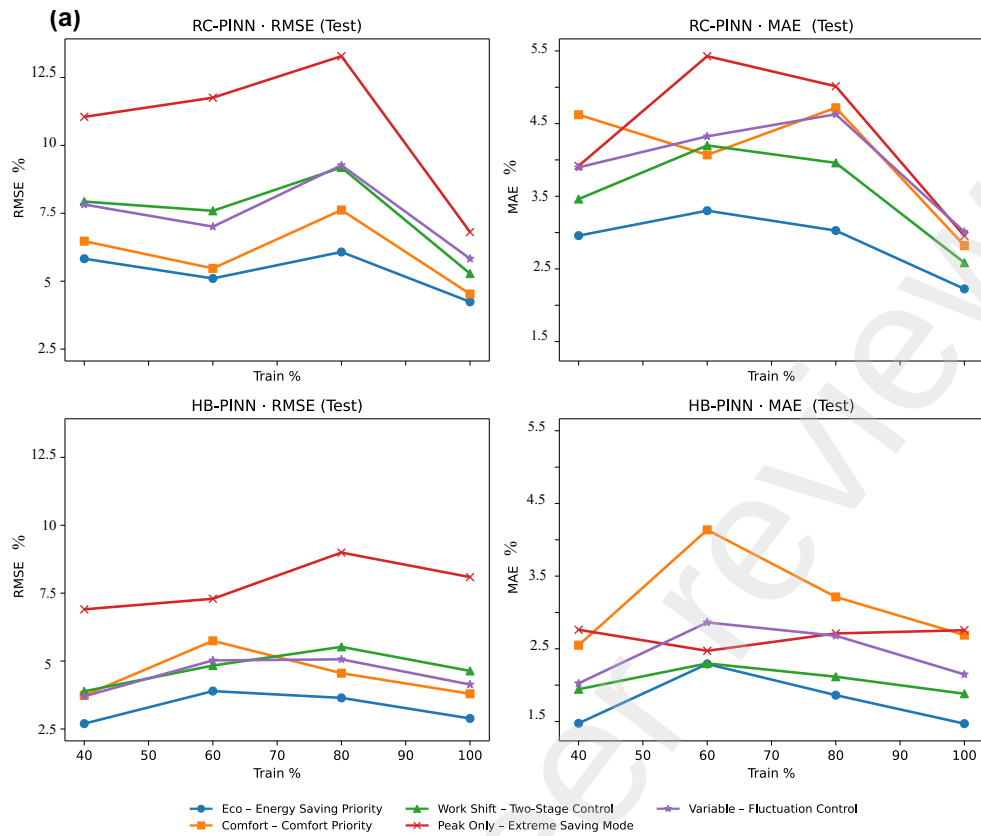
HB-PINNs address this limitation by using the full heat-balance equation. HB-PINNs achieve higher accuracy than the other models across all scenarios and data ratios. The advantage is particularly clear under the more challenging conditions. For example, in the Variable scenario with only 40% of the data used for training, HB-PINNs reach an  $R^2$  of 97.4%, whereas RC-PINNs reach 88.6%. These results suggest that incorporating the complete physics provides stronger regularisation

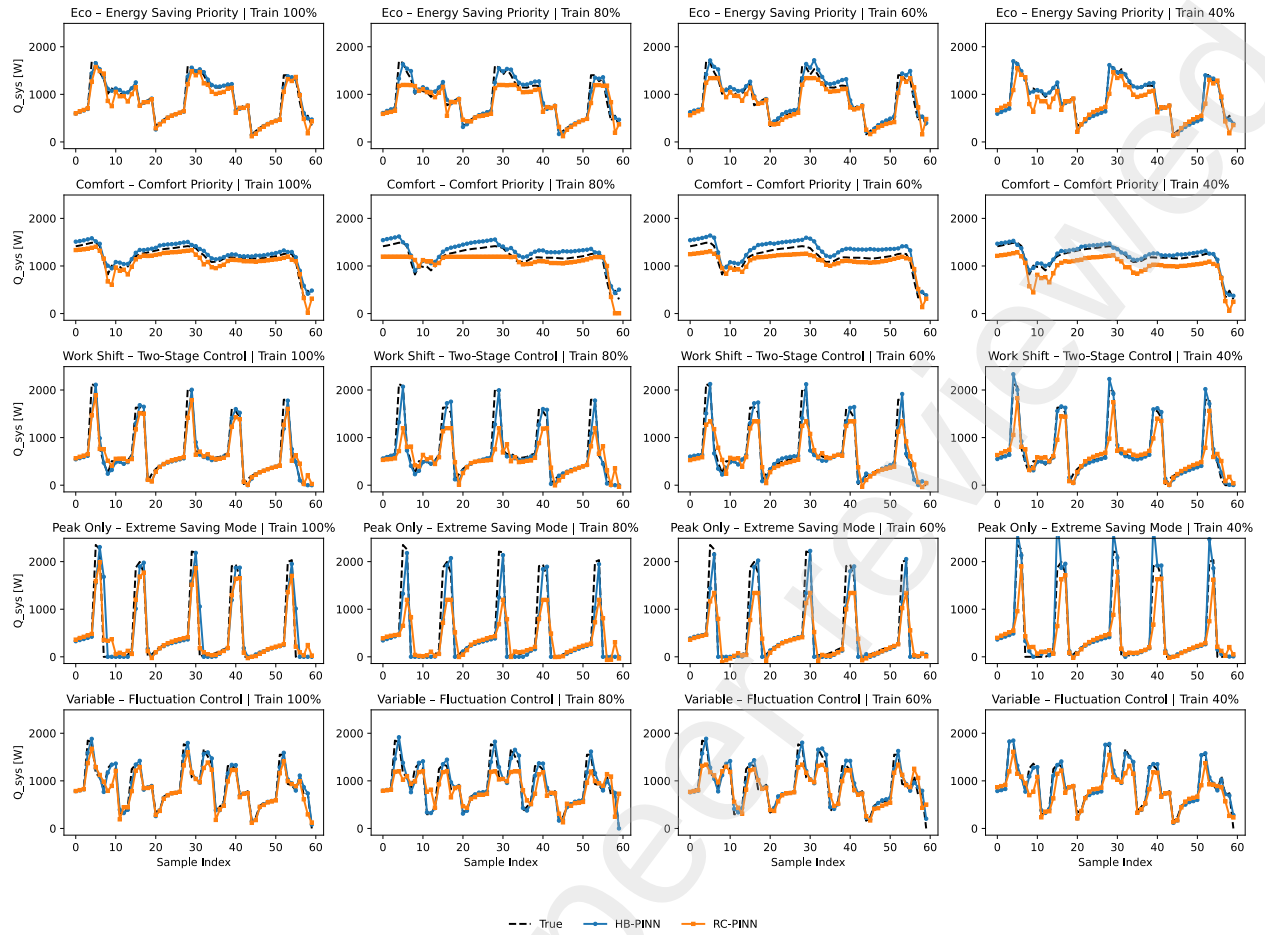
### 3.3. Comparative Visualization of RC-PINNs and HB-PINNs across Data and Scenarios

Fig. 5 shows the performance of RC-PINNs and HB-PINNs under different training ratios and test scenarios. Subfigure (a) plots both models on the same axes, allowing a direct comparison. Across most conditions, HB-PINNs achieve lower errors than RC-PINNs. Their RMSE% curves remain below those of RC-PINNs in nearly all ratios and

scenarios. For MAE, both models show higher errors when the training ratio decreases from 100% to 80%, and again from 80% to 60%. This behaviour appears in most scenarios. Small deviations can be seen in the Comfort case for RC-PINNs and in the Peak Only case for HB-PINNs. Although the trend is similar, the error levels differ. For most test scenarios, the MAE% of HB-PINNs stays below 3. For RC-PINNs, the MAE% is generally above 3, independent of the training ratio or scenario. When the training ratio falls from 60% to 40%, both models show an unexpected reduction in MAE. While Subfigure (a) presents the individual error trajectories of each model, it does not directly show the difference between them. Subfigure (b) addresses this by using heatmaps to display the absolute differences in prediction error between RC-PINNs and HB-PINNs across the training ratios and control scenarios. The columns represent the training data ratios (100%, 80%, 60%, and 40%), and the rows correspond to the five testing scenarios. Red colour indicates that HB-PINNs achieve lower RMSE or MAE than RC-PINNs, with stronger colour intensity showing a larger difference. Across most scenarios and training levels, HB-PINNs obtain lower errors. Minor exceptions appear in the Comfort scenario under the 100% and 40% training ratios, where RC-PINNs produce slightly smaller errors. In the remaining cases, HB-PINNs show clear advantages, especially under conditions with high variability such as

# HB-PINNs: Heat-Balance Physics-Informed Building Energy Forecasting



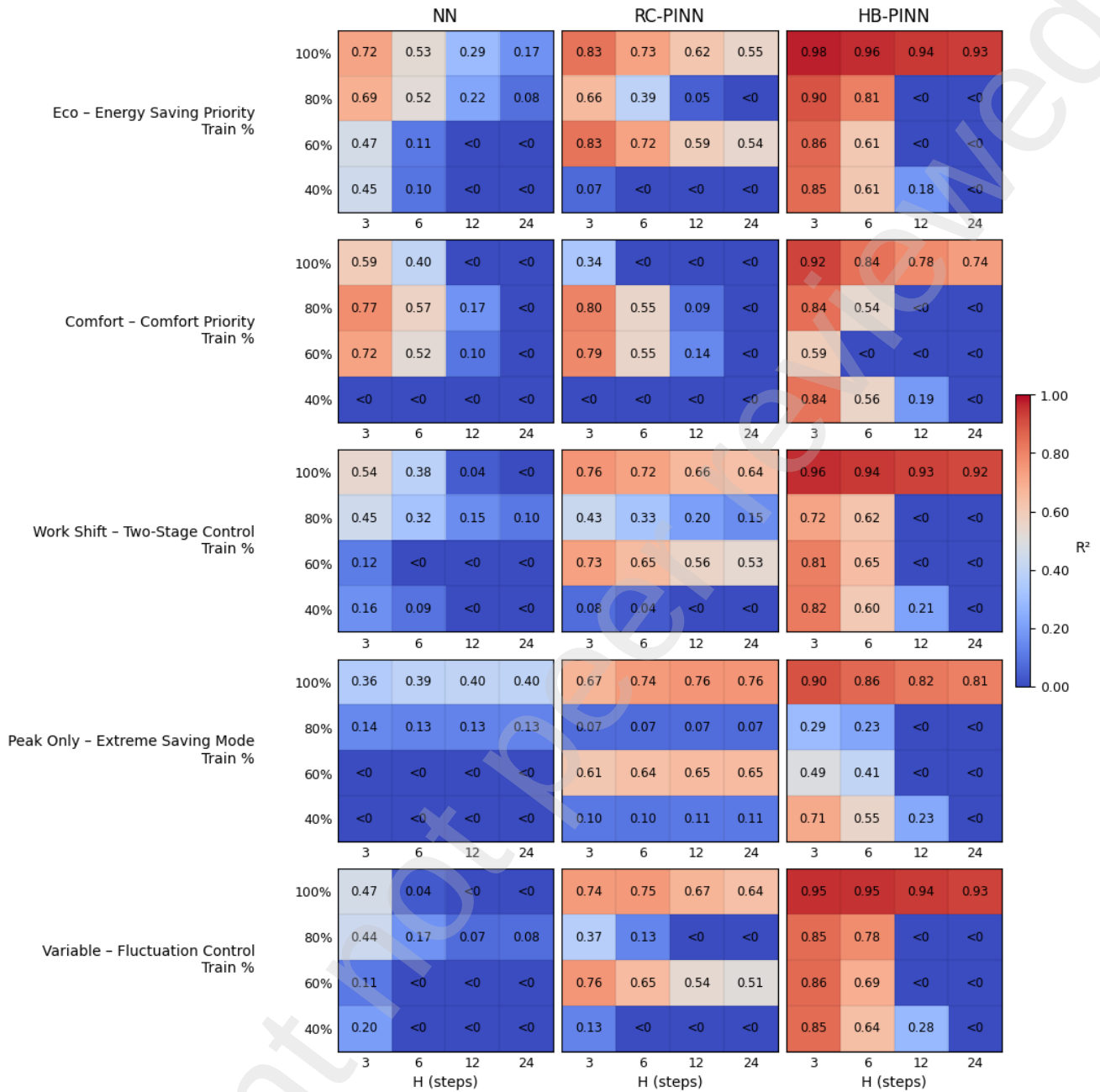


**Figure 6:** Comparison between RC-PINNs and HB-PINNs Predictions across All Five Test Scenarios under Four Levels of Training Data (5 days)

the Peak Only and Variable scenarios. The size of the improvement increases from the Work Shift to the Variable scenario, suggesting that HB-PINNs are more effective in capturing complex and dynamic thermal behaviour. The maximum observed gains reach 89.3 in RMSE and 59.1 in MAE. Subfigure (c) summarises the average RMSE and MAE across all test sets for each training ratio. The bar charts show greater stability and lower errors for HB-PINNs, particularly at the lower training ratios of 40% and 60%. These results highlight the benefit of including the full heat-balance constraint in the learning framework. Compared with the RC-based structure, the HB-PINNs formulation improves predictive accuracy and reduces overfitting, leading to better generalisation under limited data and dynamic conditions.

In addition to overall error levels, it is important to check whether each model can follow the detailed changes in heating demand. RMSE or MAE alone may miss fast changes or the basic shape of load changes. Fig. 6 compares the hourly predictions of RC-PINNs and HB-PINNs across all five scenarios and four training-data levels (5-day samples). Each subplot shows the predicted and true curves for a direct visual comparison. Across most scenarios, HB-PINNs

follow the measured load more closely. They match both the size and the general shape of the heating demand, including sharp rises and repeated cycles. In the Eco and Work Shift cases, HB-PINNs keep the timing and the size of the on/off changes, even with only 40% of the data. RC-PINNs show reduced accuracy as the training data decrease. The curves become smoother than they should be, with slower increases and lower peaks. In more dynamic cases such as Peak Only and Variable, HB-PINNs still capture steep changes and rapid movements. RC-PINNs show instability at the 60% and 40% levels and often underestimate peak loads. A small exception appears in the Comfort scenario, where HB-PINNs slightly underestimate the heating load at the 100% and 40% levels. However, at 80% and 60%, they again perform better. Overall, the figure shows that HB-PINNs can rebuild detailed heating profiles across many conditions and training levels. Their ability to keep the physical structure and maintain accuracy with limited data.

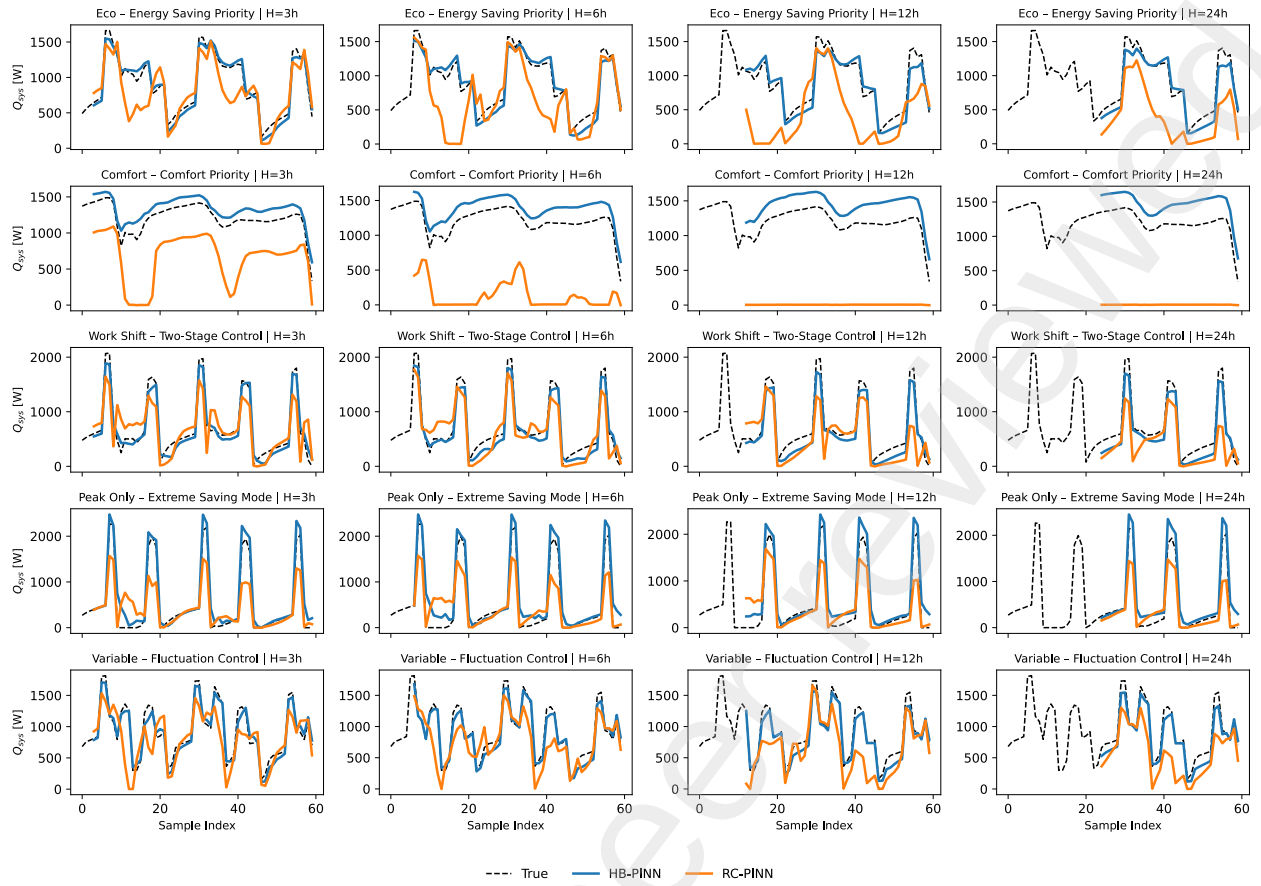


**Figure 7:** Multi-step Forecasting Performance of NN, RC-PINNs and HB-PINNs under Different Training Ratios and Forecast Horizons

### 3.4. Evaluation of Multi-Step Prediction Capability across Models

Beyond one-step-ahead prediction, multi-step forecasting is essential for proactive energy management in building systems. Accurate long-horizon forecasts enable predictive control and scheduling but are more challenging due to the compounding of uncertainties and the accumulation of errors. This section evaluates how prediction performance changes across different forecast horizons and training data availability. Three models are considered: NN, RC-PINNs,

and HB-PINNs. While Fig. 7 compares all three, Fig. 8 presents only RC-PINNs and HB-PINNs. Fig. 7 presents the multi-step forecasting performance of three models-NN, RC-PINNs, and HB-PINNs under different training data ratios (100%, 80%, 60%, 40%) and forecasting horizons (3, 6, 12, and 24 hours). The vertical axis represents five control strategy scenarios (Eco, Comfort, Work Shift, Peak Only, and Variable), while the horizontal axis denotes the prediction horizon  $H$ . The color indicates  $R^2$ , with red representing better performance and blue indicating poorer



**Figure 8:** Multi-Step Prediction of Hourly Heating Demand Profiles ( $H = 3, 6, 12, 24$  Hours) by RC-PINNs and HB-PINNs Models under Different Operation Modes with 100% Training Data

performance. When the training data are sufficient ( $\text{train}\% = 100\%$ ), the HB-PINNs model consistently achieves the highest  $R^2$  across all horizons and scenarios, with most values exceeding 0.9. The RC-PINNs model ranks second, maintaining an  $R^2$  of around 0.6–0.7 except in the Comfort scenario, where performance declines. The NN model has the lowest stability. Its ability to predict beyond the training range drops clearly as the prediction horizon increases. When the horizon is short ( $H=3$ ), HB-PINNs remain the most accurate, staying above 0.7 even when the training data are reduced. RC-PINNs show mixed behaviour as data availability decreases, without a clear trend. The NN model remains the least accurate and is the most affected by data reduction. Overall, as the amount of training data decreases and the forecasting horizon increases, the predictive performance of all three models deteriorates, which is consistent with expectations given the reduced information for training and the higher uncertainty associated with long-term forecasts.

Fig. 8 compares the actual hourly heating demand profile with the model predictions RC-PINNs and HB-PINNs. Each column corresponds to a different prediction horizon (3, 6,

12, and 24 hours). Both models adopt an autoregressive multi-step prediction mechanism, where each step prediction is recursively fed into the next. The figure displays only the hourly predictions from step 1 to  $H + 1$ , leading to blank regions at the beginning. The black dashed lines represent the measured hourly heating demand, while the coloured curves correspond to model-based predictions. For shorter horizons (e.g.,  $H = 3$  h), both models can closely follow the real heating demand profile. As the prediction horizon increases, the RC-PINNs begins to accumulate larger recursive errors due to its simplified RC formulation, leading to visible deviations during peak heating periods. In contrast, HB-PINNs use additional physical information from the full heat-balance process. This allows them to capture transient thermal behaviour more accurately and stay stable even for long forecasting horizons of up to 24 hours. This difference shows the better generalisation and robustness of the HB-PINNs structure when enough data are available.

## 4. Conclusion

This study proposed HB-PINNs framework for accurate and generalizable prediction of building heating demand under various operational strategies and levels of training data availability. By embedding the full heat balance equation into the neural networks and enabling the inference of latent physical quantities, HB-PINNs maintain physical consistency while improving data efficiency. Extensive tests across five representative control scenarios and multiple training ratios show that HB-PINNs consistently outperforms both the NN and the resistance–capacitance-based RC-PINNs. While RC-PINNs improves prediction robustness by integrating simplified physics, its performance remains constrained by the limited expressiveness of the RC model. In contrast, HB-PINNs captures more complex thermal interactions and better reconstructs high-resolution hourly load profiles, achieving higher accuracy and greater stability, particularly under dynamic conditions and data-scarce regimes. Across all five scenarios and four training ratios, the proposed HB-PINNs achieved an average  $R^2$  of 94.99%, compared with 89.96% for the RC model, 88.69% for RC-PINNs, and 68.92% for the purely data-driven NN model. In addition, HB-PINNs also performs very well in multi-step prediction tasks. HB-PINNs can always get the highest  $R^2$  values under different prediction durations (3 hours, 6 hours, 12 hours and 24 hours) and different training data ratios. When there are enough data, this value can often exceed 0.9, and even if there are fewer data, its performance is still very stable. In contrast, RC-PINNs shows a gradual loss of accuracy as the horizon extends, or the training set becomes smaller. The NN model drops off more noticeably, with lower accuracy and limited stability in longer-range predictions. These results indicate that embedding heat-balance physics into the learning process improves the generalization capability of the model. HB-PINNs maintains stable prediction accuracy across different control scenarios and prediction horizons. Its performance also remains robust under different levels of training data availability. This behaviour suggests that the proposed framework is well suited for heating demand prediction under varying operating conditions and uncertainty. A limitation of the present study is that the model is evaluated using simulation data. Future work will extend the present study by validating the proposed framework using measured building operation data collected from real-world buildings.

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